

# CadStruct-MoE: Contract-Centered Scene-Graph Understanding for Architectural CAD Drawings

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## Abstract

We present CadStruct-MoE, a contract-centered mixture-of-experts framework for converting architectural CAD floor-plan evidence into structurally valid scene graphs. Under the reviewed SVG/contract normalized-candidate evaluation boundary, the promoted P301 line reaches node macro-F1 0.962907, node accuracy 0.982498, and invalid graph rate 0.000000. Relation scoring reaches F1 0.929782 under an internal locked fine-threshold audit; this relation value is not external validation. We additionally report qualitative end-to-end scene-graph outputs and a bounded runtime raster adapter as secondary bridge evidence, while explicitly blocking external/source-heldout generalization claims until new external labels are available.

## 1 Introduction

Architectural CAD understanding requires structured reasoning over symbols, spaces, boundaries, text, and relations. CadStruct-MoE frames this task as contract-centered scene-graph understanding: normalized CAD/SVG candidates are routed to specialized experts and fused into schema-valid graphs. This formulation separates graph reasoning from raster extraction so that each result can be reported under its proper evidence boundary.

## 2 Contributions

- A contract-centered CadStruct-MoE architecture that separates normalized candidate reasoning, specialized experts, relation scoring, graph fusion, and source-boundary accounting.
- A reviewed P301 SVG/contract normalized-candidate result with node macro-F1 0.962907, node accuracy 0.982498, and invalid graph rate 0.000000.
- An internal locked fine-threshold relation audit reaching relation precision 0.994900, recall 0.872665, and F1 0.929782, reported separately from external validation.
- Qualitative end-to-end output assets from the v13 real E2E visual demo and an SVG-derived locked-smoke export path.
- A bounded secondary runtime raster adapter, frozen at P262, with overall F1 0.729861 and equipment F1 0.729524; this is bridge evidence rather than the headline claim.
- A conservative external-generalization decision: P305 blocks source-heldout generalization claims for P301, RoomSpace, SymbolFixture, and TextDimension pending new external labels.

Table 1: Reviewed P301 CadStruct-MoE main result.

| Metric             | Value    | Claim boundary                          |
|--------------------|----------|-----------------------------------------|
| Node macro-F1      | 0.962907 | SVG/contract normalized-candidate audit |
| Node accuracy      | 0.982498 | SVG/contract normalized-candidate audit |
| Relation precision | 0.994900 | internal locked fine-threshold audit    |
| Relation recall    | 0.872665 | internal locked fine-threshold audit    |
| Relation F1        | 0.929782 | internal locked fine-threshold audit    |
| Invalid graph rate | 0.000000 | SVG/contract graph-validity audit       |

Table 2: Strong and disclosure symbol classes under the reviewed SVG/contract boundary.

| Class          | F1       | Role                               |
|----------------|----------|------------------------------------|
| shower         | 0.992464 | strong recurring symbol            |
| sink           | 0.966910 | strong recurring symbol            |
| appliance      | 0.955079 | strong recurring symbol            |
| stair          | 0.926378 | strong recurring symbol            |
| column         | 0.922162 | strong recurring symbol            |
| equipment      | 0.913676 | strong recurring symbol            |
| bathtub        | 0.835616 | lower-support disclosure symbol    |
| generic_symbol | 0.784314 | residual/open-set disclosure label |

### 3 Method Overview

CadStruct-MoE separates candidate normalization, expert routing, relation scoring, and graph fusion. SVG/contract metrics evaluate normalized scene-graph reasoning, relation audits evaluate the locked thresholded relation scorer, and raster-adaptor metrics evaluate only the runtime bridge from raster-derived predictions into the contract representation. This source-boundary accounting is part of the method because it prevents SVG/contract evidence, raster evidence, and external-generalization evidence from being conflated.

### 4 Main Results

The reviewed P301 line is the current main quantitative result. It applies a conservative relation-confidence-preserved residual relabeler on top of the P297 SVG/contract candidate and changes 77 locked symbols: 54 appliance-to-equipment, 20 stair-to-equipment, and 3 column-to-equipment. This raises node macro-F1 to 0.962907 and node accuracy to 0.982498 while preserving the P297 relation score. Relation F1 is 0.929782 under the internal locked fine-threshold audit at threshold 0.985, with relation precision 0.994900 and recall 0.872665. This relation audit is useful for the paper but must not be described as external validation.

Per-class symbol results show the strongest recurring categories above 0.90 F1: shower 0.992464, sink 0.966910, appliance 0.955079, stair 0.926378, column 0.922162, and equipment 0.913676. Bathtub reaches 0.835616. The remaining weak disclosure category is generic\_symbol at 0.784314, which should be framed as a residual/open-set label rather than a solved recurring fixture category.

Table 3: End-to-end output inventory and paper use.

| Artifact                              | Role                             | Use in paper                   |
|---------------------------------------|----------------------------------|--------------------------------|
| v13 real E2E visual demo              | qualitative E2E visualization    | main qualitative figure        |
| model-fused locked-smoke export       | SVG-derived E2E export smoke     | export-chain demonstration     |
| scene-graph locked-smoke sanity check | contract/export sanity check     | appendix or internal note only |
| image-only MoE v14                    | negative image-only E2E evidence | limitations/appendix           |
| P263/P262 raster adapter              | secondary runtime raster bridge  | bounded bridge paragraph       |

## 5 End-to-End Outputs

The paper can include a bounded end-to-end output subsection. The main qualitative figure source is `reports/vlm/visual_demo_model_v13_real_e2e`, which contains five CubiCasa visual-demo cases with input images, overlays, side-by-side outputs, per-sample summaries, a review pack, and a paper-candidate figure. These figures demonstrate the system output format and scene-graph visualization, but they are not source-heldout or pure-raster evidence.

For an SVG-derived export-chain demonstration, `model-fused locked-smoke eval` runs registered expert checkpoints on SVG-derived candidates across 64 cases. It reports node F1 0.723279 and relation F1 1.000000, with family node F1 of 0.774974 for boundary, 0.886297 for space, 0.456018 for symbol, and 0.994737 for text. This path is useful as an end-to-end smoke/export demonstration, not as the headline model result.

The perfect-score scene-graph locked-smoke sanity check should only be cited as a contract/export sanity check because its behavior is gold-compatible and should not be presented as model recognition performance. The image-only path remains negative evidence: it passes source-integrity checks on 16 rows, but proposal mean F1 is 0.0, the final scene graph has 17 nodes and 0 edges, and the artifact is explicitly not adopted.

## 6 Secondary Runtime Raster Bridge

The runtime raster adapter is secondary. P262 improves the promoted P232 raster baseline from F1 0.726314 to 0.729861, with equipment F1 0.729524, but this remains bridge evidence. The manuscript should not frame the project as a pure raster detector or transfer SVG/contract metrics into runtime-raster claims.

## 7 External Generalization and Limitations

external/source-heldout generalization remains blocked by P305. The only true source-heldout expert evaluated in P305b is WallOpening, and transfer is weak: `cvc_fp` to floorplancad macro-F1 is 0.485651, while floorplancad to `cvc_fp` macro-F1 is 0.159075. RoomSpace, SymbolFixture, and TextDimension are blocked by single locked-source coverage or missing external labels. The relation heldout scorer is internal record-heldout within the locked benchmark and must not be used as external source validation.

## 8 Figure Plan

The main qualitative figure should use `figures/paper_candidate_figure_v13_real_e2e.png`. Its caption should state that it is a qualitative end-to-end scene-graph output, not source-heldout or

Table 4: P305 external-generalization decision carried into the manuscript.

| Area                        | Status                       | Paper wording                                     |
|-----------------------------|------------------------------|---------------------------------------------------|
| P301 external validation    | blocked                      | do not describe as external validation            |
| WallOpening source transfer | evaluated but weak           | zero-shot source transfer not supported           |
| RoomSpace                   | blocked                      | needs non-CubiCasa locked room labels             |
| SymbolFixture               | blocked                      | needs FloorPlanCAD/internal symbol fixture labels |
| TextDimension               | blocked                      | needs FloorPlanCAD/internal OCR/text labels       |
| Relation heldout scorer     | internal record-heldout only | not external source validation                    |

pure-raster evaluation. A second schematic can show the claim-boundary ledger: P301 metrics evaluate reviewed SVG/contract normalized-candidate recognition; relation F1 is an internal locked fine-threshold audit; P262 is bounded runtime raster bridge evidence; P305 keeps source-heldout generalization claims blocked until new external labels are available.

## 9 Conclusion

CadStruct-MoE should be read as a contract-centered scene-graph reasoning contribution with explicit source boundaries. The reviewed SVG/contract P301 result provides the main evidence for expert routing and graph fusion, while the end-to-end visual output, SVG-derived export smoke path, image-only negative evidence, and frozen P262 raster adapter provide bounded supporting context.